SMART INDIA HACKATHON 2026

SHADOWNET FORESIGHT

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Problem Statement ID – 26153
Problem Statement Title – AI based Network Attack Forecasting from Network Traffic Data
Theme – Blockchain & Cybersecurity
PS Category – Software



IDEA TITLE

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Proposed Solution — ShadowNet ForeSight

• Forecasts attack risk and likely stage at +5, +10 and +15 seconds from the latest 60 seconds of traffic.

• Unifies PCAP, Cowrie, NetFlow/IPFIX and Kafka events into leakage-safe five-second network states.

• Combines calibrated risk, unknown-network rejection, Integrated Gradients and defence simulation in one SOC dashboard.

TECHNICAL APPROACH

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Traffic → States → Forecast → Decision

• 15–21 flow and host-graph features × 12 historical windows feed a multi-head temporal model.
• LSTM / GRU / TCN / Transformer benchmark; promoted checkpoint selected only on the validation day.
• Per-horizon temperature calibration + F1 threshold search under a 5% validation FPR budget.
• FastAPI, PyTorch, scikit-learn, Scapy, WebSockets, Kafka and a responsive JavaScript dashboard.

FEASIBILITY AND VIABILITY

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Working prototype with explicit safety gates

• Implemented: PCAP replay, streaming telemetry, live forecasting, explanations, RBAC, audit logs and mitigation simulation.
• Risk: low recall and strong cross-network shift. Response: restricted claims, target-network calibration and OOD alert suppression.
• Risk: identity fields are absent from CIC training. Response: unsupported features are masked until labelled honeypot retraining.
• Deployment path: collect local benign traffic → validate/calibrate → shadow SOC mode → analyst-approved actions.

IMPACT AND BENEFITS

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Earlier, safer SOC decisions—not autonomous blocking

• Forecasted 14 of 215 attack onsets early on the untouched CIC test day; median successful lead time: 15 seconds.
• Fixed-test F1: 5.2% / 7.5% / 6.1% at +5 / +10 / +15 seconds; false-positive rate: 1.8%–3.3%.
• CTU-13 raw FPR is 100%, but 100% is detected as out-of-distribution and its automated alerts are suppressed.
• Benefits: faster triage, transparent evidence, safer unknown-network handling and auditable response planning.

RESEARCH AND REFERENCES

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Research foundations and reproducible evidence

- CSE-CIC-IDS2018 — seven whole-day train/validation/test partitions; no sequence crosses a day boundary.
- CTU-13 Scenario 11 — external diagnostic only; never used for model or threshold selection.
- IEEE SRDS 2025 Cowrie sample — 300 real records → 281 states with zero unknown endpoints.
- Methods: multi-horizon temporal learning, temperature calibration, OOD rejection and Integrated Gradients.
- Evidence: PHASES.md and fixed evaluation JSON reports in the project repository.